

MAI1202 Assignment #1

The work that you complete and submit must be your own, not the work of others, nor the creation of an AI tool. It is permissible to discuss the assignment with others in general terms, discuss general approaches to problems, help someone to understand a concept or give them a bit of advice if they have a problem, etc. However, you are not permitted to ‘work together’, nor to give/receive solutions (whether partial or complete) to/from others.

If you are unclear at all about what is permissible, please ask your professor.

Late submissions will not be accepted without prior arrangement.

Scenario

- You will consider a data set that will be provided by your professor with the assignment.

Task

Your task is to build and compare a variety of classification or regression models for the given scenario. More specifically, you will be experimenting with deep network designs and techniques; you will also be comparing the performance and computational requirements of these networks with ‘traditional’ ML models and ensemble methods.

1. Build a deep neural network of your own design to make predictions, but with two versions:
 - a. In one version, do not apply any techniques to improve generalization.
 - b. In the second version apply appropriate generalization techniques.
 - c. Compare the performance and training time of these two versions.
2. Build a model with a different design (applying appropriate generalization techniques).
 - a. Compare its performance/training time to your first model.
3. In addition to these deep models, build:
 - a. A ‘conventional’ machine learning model
 - b. An ensemble model
4. Compare the performance/training time for these models to the deep models.
5. Reflect on the results you obtained: explain the relative performance of the models you tested on this problem and dataset.

Requirements

Your submission should consist of a single notebook file, containing both your report text and your code/output. (The two should be integrated appropriately; do not put all of the text at the beginning/end.)

Submit both your Notebook file, as well as a PDF file of your notebook.

Important note: You may be interviewed or quizzed about your submission, to ensure your understanding of the work submitted. Your ability to demonstrate understanding will be factored into your grade.